Curve Sketching

Example 1: Sketch a~~nd~~ properly labelled graph of $f(x)=\dfrac{x}{(3x+1)^2}$, given

$$f'(x)=\frac{-3x+1}{(3x+1)^3} \text{ and } f''(x)=\frac{6(3x-2)}{(3x+1)^4}.$$

* Domain: $\left(-\infty,-\frac{1}{3}\right)\cup\left(-\frac{1}{3},\infty\right)$

* NO HOLES

* V.A. @ $x=-\frac{1}{3}$

* x & y-int

→ x-int: set $y=0$

$\Rightarrow x=0$

$\therefore$ x-int @ $(0,0)$

→ y-int: set $x=0$

$\Rightarrow y=0$

$\therefore$ y-int @ $(0,0)$

* Behaviour Near V.A.

$$\lim_{x\to -\frac{1}{3}^-}\frac{x}{(3x+1)^2}=-\infty \qquad \lim_{x\to -\frac{1}{3}^+}\frac{x}{(3x+1)^2}=-\infty$$

* H.A.

H.A. @ $y=0$

→ $f(100)=\dfrac{100}{(301)^2}>0 \Rightarrow$ above H.A. as $x\to\infty$

→ $f(-100)=\dfrac{-100}{(-299)^2}<0 \Rightarrow$ below H.A. as $x\to-\infty$

H.A. Cross test

$\dfrac{x}{(3x+1)^2}=0 \Rightarrow x=0$ $\therefore$ crosses H.A. @ $(0,0)$

* Local Extrema

critical # is $x = \frac{1}{3}$ * $x = \frac{-1}{3}$ (V.A.)

Interval	$(-\infty, -\frac{1}{3})$	$(\frac{-1}{3}, \frac{1}{3})$	$(\frac{1}{3}, \infty)$
$-3x+1$	+	+	−
$(3x+1)^3$	−	+	+
$f'(x)$	−	+	−
$f(x)$	↓	↑	↓

V.A. @ $x = -\frac{1}{3}$

Max @ $x = \frac{1}{3}$ pt. $\left(\frac{1}{3}, \frac{1}{12}\right)$

* Concavity & Possible Poi

$f''\left(\frac{2}{3}\right) = 0$; $f''\left(\frac{-1}{3}\right)$ DNE (V.A.)

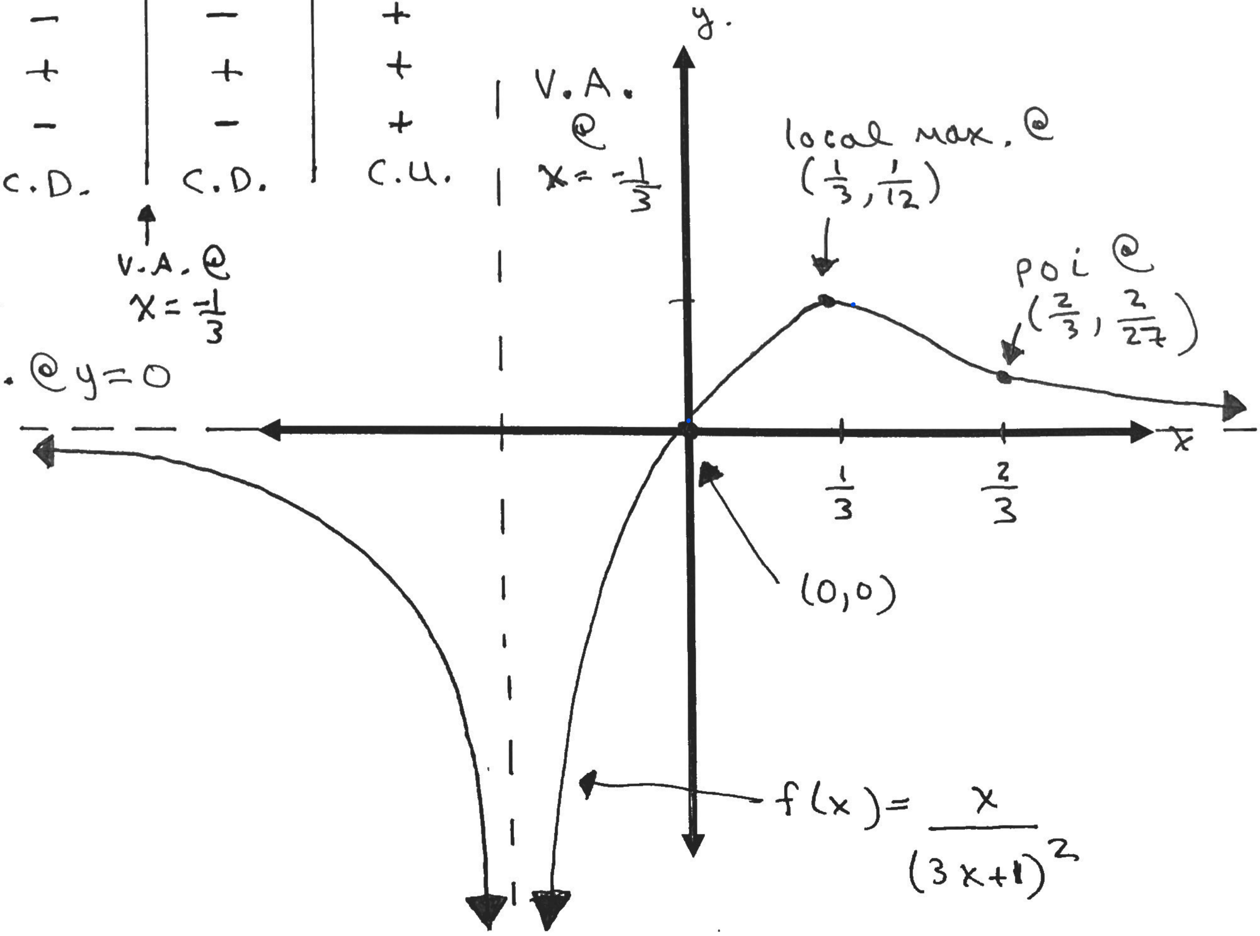
$x = \frac{2}{3}$ and $x = \frac{-1}{3}$ (V.A.)

Interval	$(-\infty, -\frac{1}{3})$	$(\frac{-1}{3}, \frac{2}{3})$	$(\frac{2}{3}, \infty)$
$3x-2$	−	−	+
$(3x+1)^4$	+	+	+
$f''(x)$	−	−	+
$f(x)$	C.D.	C.D.	C.U.

V.A. @ $x = \frac{-1}{3}$

$\therefore$ poi @ $\left(\frac{2}{3}, \frac{2}{27}\right)$

H.A. @ $y = 0$



Example 2: Sketch and properly labelled graph of $f(x) = -3x^{\frac{5}{3}} + 15x^{\frac{2}{3}}$, given

$$f'(x) = -5x^{-\frac{1}{3}}(x-2) \text{ and } f''(x) = -\frac{10}{3}x^{-\frac{4}{3}}(x+1).$$

$= -3x^{\frac{2}{3}}(x-5)$

$\uparrow$ $x=0$ $\quad$ $\uparrow$ $x=5$

* Domain: $(-\infty, \infty)$

* NO HOLES

* NO V.A.
* NO H.A. / NO O.A.
* x & y-int

→ x-int: set $y=0$

$\Rightarrow x = 0, 5$

$\therefore$ x-int @ $(0,0)$ & $(5,0)$

→ y-int: set $x=0$

$\Rightarrow y = 0$

$\therefore$ y-int @ $(0,0)$

* Local Extrema

critical #'s are $x = 0, 2$

Interval	$(-\infty, 0)$	$(0,2)$	$(2, \infty)$
-5	$-$	$-$	$-$
$x^{\frac{1}{3}}$	$-$	$+$	$+$
$x-2$	$-$	$-$	$+$
$f'(x)$	$-$	$+$	$-$
$f(x)$	$\downarrow$	$\uparrow$	$\downarrow$

Min @ $(0,0)$ $\quad$ Max @ $(2, 14.29)$

* Concavity & possible poi

$f''(0)$ DNE ; $f''(-1) = 0$

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, \infty)$
$-\frac{10}{3}$	$-$	$-$	$-$
$x^{4/3}$	$+$	$+$	$+$
$x+1$	$-$	$+$	$+$
$f''(x)$	$+$	$-$	$-$
$f(x)$	C.U.	C.D.	C.D.

poi @ $(-1, 18.)$

No poi @ $(0,0)$ "cusp"